SECTION 265568
ATHLETIC FIELD LIGHTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General, Special and Supplementary Conditions, Division 01 Specification Sections, and Section 260010 “General Requirements for Electrical Systems” apply to this Section.

1.2 SUMMARY

A. DescriptionThis project consists of retrofitting athletic lighting systems at the five Lewisville ISD High Schools in Texas: **Flower Mound High School (Town of Flower Mound), Hebron High School (Carrollton), Lewisville High School (Lewisville), Marcus High School (Town of Flower Mound), and The Colony High School (The Colony)**. Contractor shall supply and install a complete LED lighting system that meets or exceeds the quantity, quality, and operational criteria for the following outdoor sports venues as outlined in the contract documents and University Interscholastic League (UIL) standards. Contractor shall remove and replace existing luminaires with new LED fixtures, reuse existing poles, foundations, and electrical circuits from the source breaker to the poles, and incorporate theatrical RGB effects and mandatory mobile app controls. Contractor shall ensure standardization of luminaires, controls, and components across all five high schools to facilitate maintenance, reduce inequity, and support staff mobility between campuses.

1. **Football Fields.**
2. **Track.**
3. **Football Bleachers (Home and Visitor)**

B. The primary goals of the system are as follows:

1. Guaranteed Light Levels: Provide guaranteed light levels to ensure optimum athletic performance conditions and ensure safety for players and officials during competition.
2. Environmental Light Control: Minimize spill light to adjoining properties and glare to the players, spectators, and neighbors. **Design must comply with City/Town ordinances.**
3. Energy Efficiency: Minimize energy costs for operating the system.
4. Maintenance and Reliability: Eliminate all maintenance costs for the duration of the system warranty.
5. Control and Monitoring: Provide a remote on/off control system for the lighting system with mandatory mobile app integration (compatible with iOS and Android devices) for scheduling, monitoring, and operation, including theatrical RGB show lights with at least five owner-determined presets. Proactively monitor the system to detect luminaire outages for the duration of the system warranty. Include all communication, infrastructure, and monitoring costs.

C. Section Includes:

1. Luminaires.
2. Poles and support structures.
3. Power distribution and controls.

D. Delegated Design and Turn-Key Delivery: Contractor shall use a delegated design approach, engaging a qualified, Texas-licensed professional engineer to analyze existing poles and foundations, design attachments or reinforcements, and certify compliance with codes and loads. Contractor shall deliver a turn-key project, assuming full responsibility for engineering, procurement, installation, integration, testing, and commissioning, including all coordination, permitting, and compliance with jurisdictional requirements (Lewisville, Town of Flower Mound, The Colony, and Carrollton). Contractor shall account for site-specific conditions and ensure a fully operational system upon substantial completion, without additional Owner intervention post-award.

E. Bidder Responsibilities and Pre-Bid Due Diligence: Contractor shall conduct thorough field condition assessments prior to bid submission, including site visits to evaluate existing components for anomalies, damage, or non-compliance. Contractor shall include costs for potential repairs, reinforcements, breaker replacements, wiring modifications, and testing in base bid pricing. Failure to identify and price these during pre-bid will not entitle Contractor to additional compensation. Submittals shall include a structural analysis report and certification of system soundness.

F. Scope per School: The project includes five (5) separate high school athletic lighting retrofits. All requirements in this Section (including, but not limited to, preinstallation meetings, field testing and photometric verification, training, demonstration, drone documentation, manufacturer/vendor site visits and meetings, and warranties) shall be furnished fully and separately at each school. Quantities stated in hours, visits, sessions, or documentation shall be interpreted as the minimum required per school unless otherwise noted.

1.3 REFERENCES

A. Abbreviations and Acronyms:

1. EPA: Equivalent projected area.
2. LC: Lighting Certified.
3. LLF: Light Loss Factor.
4. TLCI: Television Lighting Consistency Index.

B. Definitions:

1. Coefficient of Variation (CV): A statistical measure of the weighted average of all relevant illumination values for the playing area, expressed as the ratio of the standard deviation for all illuminance values to the mean illuminance value.
2. Illuminance: The metric most commonly used to evaluate lighting systems. It is the density of luminous flux, or flow of light, reaching a surface divided by the area of that surface.

   a. Horizontal Illuminance: Measurement in foot-candles, on a horizontal surface 36 inches above ground unless otherwise indicated.

b. Target Illuminance: Average maintained illuminance, calculated by multiplying initial illuminance by LLF. System shall not fall below this level throughout the system life cycle.
c. Vertical Illuminance: Measurement in foot-candles, in two directions on a vertical surface, at an elevation coinciding with plane height of horizontal measurements unless noted otherwise.

3. Lumen Maintenance: Projections per IES TM-21-21 to ensure target levels throughout warranty.
4. Measurements: Per IES LM-79-19 for photometric testingGlare Rating (GR): an indication of glare experienced by the observer for each point in on a calculation grid as defined by CIE document 112-1994.
5. Light-Loss Factor (LLF): A factor used in calculating the level of illumination after a given period of time and under given conditions. It takes into account temperature, dirt accumulation on the luminaire, lamp depreciation, maintenance procedures, and atmospheric conditions. An LLF includes a recoverable light-loss factor.
6. Uniformity Gradient (UG): The rate of change of illuminance on the playing field, expressed as a ratio between the illuminances of adjacent measuring points on a uniform grid.
7. Uniformity Ratio (UR): A measure for the quality of the illumination, defined by a ratio of the highest and lowest calculated or measured illuminance values (Max:/Min).

C. Reference Standards: The following publications are referred to in the text by the basic designation only. The edition/revision of the referenced publications shall be the latest date as of the date of the Contract Documents, unless otherwise specified.

1. American Association of State Highway and Transportation Officials (AASHTO)
   a. AASHTO LTS, Standard Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals.
2. American Society of Civil Engineers (ASCE): ASCE 7, Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
3. ASTM International:
   a. ASTM A123, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
   b. ASTM A572, Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
   c. ASTM A615, Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
   d. ASTM A416, Standard Specification for Low-Relaxation, Seven-Wire Steel Strand for Prestressed Concrete.
   e. ASTM B240, Standard Specification for Zinc and Zinc-Aluminum (ZA) Alloys in Ingot Form for Foundry and Die Castings.
   f. ASTM C1089, Standard Specification for Spun Cast Prestressed Concrete Poles.
4. Illumination Engineering Society (IES):
   a. IES RP-6-24 (2024), Sports, and Recreational Area Lighting.
   b. IES TM-21-21 (2021), Projecting Long-Term Luminous, Photon, and Radiant Flux Maintenance of LED Light Sources.
   c. IES LM-5-04 (2004), Guide for Photometric Measurements of Area and Sports Lighting Installations.

5. International Dark Sky Association (IDA):
   a. IDA-Criteria for Community-Friendly Outdoor Sports Lighting.
6. University Interscholastic League (UIL) .

1.4 COORDINATION

A. Coordinate pole layouts and installation of luminaires with other trades.

1.5 PREINSTALLATION MEETING

A. Schedule preinstallation meeting with Architect, Engineer, Owner, Manufacturer's Representative and all affected trades. **Separate preinstallation meetings shall be scheduled for each school unless the owner agrees in writing to combine meetings.** Agenda topics should include, but are not limited to, the following:

1. Methods and procedures related to product installation.
2. System installation schedule and material availability.
3. Planned conductor routing.
4. Planned building enclosure penetrations.
5. Planned attachment points.
6. Building material compatibility.
7. Cutting and patching requirements.
8. Lighting control schedules and programming requirements.

1.6 SUBMITTALS

A. Product Data: Submit for each type of product. Arrange in order of luminaire designation and include data on features, accessories, finishes, and the following:

1. Physical description and dimensions of the luminaires.
2. LED driver data, including UL listing and recognition, ANSI certification, and energy efficiency data.
3. LED source data, including life and output (lumens, CCT, and CRI).
4. Photometric data and adjustment factors based on laboratory tests, complying with IES "Lighting Measurements Testing and Calculation Guides," of each lighting luminaire type.
   a. Manufacturer Certified Data: Provide photometric data certified by manufacturer's laboratory with a current accreditation under the National Voluntary Laboratory Accreditation Program (NVLAP) for Energy Efficient Lighting Products..
   b. Include a summary of the horizontal and vertical aiming angles for each luminaire type.
   c. Include projected lumen maintenance per IES TM-21-21 and indicate the adjustment factors for the luminaire as applied in this Project.
5. Means of attaching luminaires to supports and indication that attachment is suitable for components involved.
6. Effective projected area (EPA) of luminaires, supports, and accessories.

B. Shop Drawings: Provide the following

1. Anchor bolt templates keyed to specific poles and certified by manufacturer.
2. Method and procedure of pole installation. Include manufacturer's written installations.
3. Include details of luminaire assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
4. Include diagrams for power, signal, and control wiring.
5. Include installation and construction details.

C. Delegated Design Submittals: Comply with performance requirements and design criteria including analysis data signed and sealed by the qualified professional responsible for their preparation.

1. Lighting system design analysis and calculations:

   a. Include point calculations of horizontal and vertical illuminance, UR, UG, CV, and GR at each grid point for grid size and spacing indicated.
   b. Include point calculation of horizontal and vertical illuminance at each calculation point in spill light areas.
   c. Calculations of source intensity of luminaires observed at eye level from indicated properties near the playing fields.
   d. Include LLF used in each design calculation.

2. Electrical system design analysis and calculations:

   a. Total connected and estimated peak-demand electrical load, in kilowatts, of lighting system.
   b. Capacity of feeders required to supply lighting system.

3. Structural analysis, foundation design and pole selection:

   a. Include material descriptions, dimensions, anchor bolts, support devices, and calculations.
   b. Include plans, elevations, sections, and details for foundations, luminaire mounting, and pole attachments.
   c. Include the moment, shear (horizontal) force, and axial (vertical) force at ground level for each pole
   d. Include means of attaching luminaires to supports and indication that attachment is suitable for components involved.

D. Pole and Support Component Certificates: Signed by pole manufacturer, certifying that products are designed for indicated load requirements according to AASHTO LTS and that load imposed by luminaire and attachments has been included in design. The certification shall be based on design calculations signed and sealed by a professional engineer.

E. Qualification Data: Manufacturer certifications for qualified installer.

F. Sample Sysem Warranties.

G. Test and Inspection Reports

1. Test procedures and equipment used.
2. Graphical or numerical test results that show conformance with requirements.

3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

H. Closeout Submittals

1. In addition to items specified in Division 01 and Section 260010 "General Requirements for Electrical Systems", include the following:

   a. Preventative Maintenance Warranty
2. Drone documentation footage and files as specified in Section 3.6.
3. **Written photometric verification and compliance certification from the lighting manufacturer's authorized representative for each school.**

1.7 QUALITY ASSURANCE

A. Manufacturer Qualifications:

1. Five continuous years, minimum, design and manufacture of the materials and equipment for professional sports lighting installations.
2. Manufacturer's primary business includes fabricating sports lighting and providing professional engineering services needed to assume engineering responsibility.
3. Maintain, within 100 miles of Project site, a service center capable of providing training, parts, maintenance, emergency repairs to equipment, and response times less than eight hours.
4. Employ trained staff that are available 24/7 to provide scheduling support and assist with reporting needs.

B. Engineering Responsibility: Preparation of delegated-design submittals and comprehensive engineering analysis shall be performed under the supervision of a qualified lighting designer.

1. The lighting designer shall be certified as an LC by the National Council on Qualifications for the Lighting Professions (NCQLP).

C. Installer Qualifications:

1. Manufacturer's authorized representative who is trained and certified by the manufacturer for installation of sports lighting systems.
2. Minimum of five years successful installation experience with Projects of similar size and complexity.
3. Upon request, be able to provide a minimum of three reference accounts at which similar work, both in scope and complexity, have been completed by The Contractor within the last three years.

D. Luminaire Photometric Data Testing Laboratory Qualifications: Luminaire manufacturers' laboratory accredited under the NVLAP for Energy Efficient Lighting Products.

E. Welding Qualifications: Qualify procedures and personnel according to the following:
1. AWS D1.1/D1.1M.
2. AWS D1.2/D1.2M.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Store poles on decay-resistant skids at least 12 inches above grade and vegetation. Support poles to prevent distortion and arrange to provide free air circulation.

B. Retain factory-applied pole wrappings on metal poles until right before pole installation. Handle poles with web fabric straps.

C. Protect mechanical finishes on exposed surfaces from damage by providing a factory applied, field removable temporary protective covering.

1.9 WARRANTY

A. Manufacturer Warranty: Manufacturer agrees to repair or replace system athletic lighting system and components that fail in materials, structural integrity, or workmanship within minimum 10 years from date of Substantial Completion.

1. Manufacturer may exclude lightning damage, hail damage, vandalism, abuse, or unauthorized repairs from warranty period.
2. **Add Alternate:** Provide additional alternate cost for extending warranty from 10 years to 25 years.

B. System Performance Warranty: Supply a signed warranty guaranteeing system performance, including any labor and parts necessary to restore the operation of the system to original design criteria for a performance period of 10 years from Substantial Completion.

1. The manufacturer shall guarantee the specified illumination levels, uniformity, obtrusive light control indicated in the system performance criteria for the duration of the warranty period.
2. All outages shall be immediately repaired when the usage of the field is materially impacted.
3. The manufacturer agrees to repair or replace components of luminaires, luminaire supports, and luminaire alignment products and to correct misalignment that occurs after successful acceptance tests for the duration of the warranty period.
4. Manufacturer shall maintain specifically funded financial reserves to assure fulfillment of the warranty for the full term.
5. Manufacturer may exclude weather conditions such as lightning or hail damage, vandalism, abuse, and unauthorized repairs and alterations from special warranty coverage.
6. **Add Alternate:** Provide additional alternate cost for extending warranty from 10 years to 25 years.

C. Preventative Maintenance: Manufacturer shall remotely monitor the performance of the lighting system, including on/off status, hours of usage, and luminaire outages for a period of 10 years from substantial completion.

1. Manufacturer shall provide all preventative and spot maintenance, including parts and labor for the duration of the warranty period.
2. Individual luminaire outages shall be repaired when the usage of the field is materially impacted.

3. Manufacturer maintain, within 100 miles of Project site, a service center capable of providing training, parts, maintenance, emergency repairs to equipment, and response times less than eight hours.
4. **Add Alternate:** Provide additional alternate cost for extending warranty from 10 years to 25 years.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency acceptable to the authority having jurisdiction and marked for intended location and application.

B. Source limitations: Design, manufacture, test, and certify all components to operate as a complete system. Components include all luminaires, harnesses, drivers, and enclosures.

C. All luminaires, visors, and support assemblies shall withstand 150 mph winds and maintain luminaires aiming alignment.

D. Provide RoHS compliant products.

2.2 SYSTEM PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, to design lighting and power systems.

B. Illuminance Criteria: Maintain the following values throughout the system performance warranty period. To limit over-lighting, the average target illuminance may not exceed 10 percent above the listed values.

Sport	**Lighting Area**	**Target Horizontal Illuminance (avg)**	**Uniformity**		
			UR	**UG**	**CV**
Football/Soccer	Field White Lighting	75 fc	≤ 2.0:1	≤ 1.5	≤ 0.21
	Bleachers (Home & Visitor) White Light*	15 fc	N/A	N/A	N/A
	Field & Bleachers (Home & Visitor) Color, RGB Light	2 fc	≤ 3.0:1	≤ 1.5	≤ 0.25
Track	Track Oval	30 fc	≤ 4.0:1	≤ 2.0	≤ 0.30

*Refer to Emergency Egress Lighting Requirements in this specification.

C. Illumination Calculations: Computer-analyzed point method complying with IES RP-6 to optimize selection, location, and aiming of luminaires.

1. Grid Pattern Dimensions: Lighting calculations shall be developed, and field measurements taken using number of points and grid spacing indicated, centered on playing surface.
2. Light Loss Factor: A light loss factor commensurate with the fixture manufacturer's TM-21 light loss data shall be applied to the initial illuminance calculations.

Sport	Min # of Grid Points	Grid Spacing
Football (White Light)	72	30 ft by 30 ft
Football Field & Bleachers (Home & Visitor) Color, RGB	72	30 ft by 30 ft
Track (White Light)	46	20 ft by 20 ft

D. Hours of Usage: Guaranteed design shall be based on annual usage of 800 hours.

E. Shadow Control: Utilize repetitive aiming from different mounting locations to limit hard line shadows created by the players.

1. Divide the playing area into three equal zones.
2. Aiming in each zone shall have a minimum of three overlapping lighting arrays per side.

F. Discomfort Glare: Minimize direct glare for primary players and spectators during standard game mode operation as well as during theatrical effects.

1. GR ≤ 40.

G. Flicker: 5 percent maximum.

H. Obtrusive light control: minimize impact on adjacent properties and surrounding environment, comply with IDA criteria for Community Friendly Sports Lighting and the following requirements:

1. Spill Light:
   a. No more than 15-percent of the total system designed lumens may be outside the region defined by a 33 foot offset around the playing field boundary.
2. Light Trespass:
   a. Maximum horizontal illuminance (fc) at property line: 0.00 fc
   b. Maximum vertical illuminance (fc) at property line: 0.00 fc
3. Uplight:
   a. No more than 8-percent of the total directly applied system lumens may be emitted above the 90-degree horizontal.
4. Glare:
   a. Maximum luminous intensity from any luminaire at any viewing angle along the property line, measured 5 feet above grade at 10 foot spacing: 1000 cd

I. Luminaire Mounting, Aiming, and Placement: Comply with IES RP-6-24, with consideration for uniformity and glare requirements.

1.

2.3 LUMINAIRES AND DRIVERS

A. Description: LED luminaire with rugged, weather-tight housing, suitable for installation in wet locations.

B. Manufacturers: Specifications are based on open criteria to allow competitive bidding from multiple manufacturers. Acceptable manufacturers subject to compliance with requirements, provide products by one of the following:

1. Musco
2. Ephesus
3. M3 Innovation Sports Lighting
4. Qualite Sports Lighting
5. Techline Sports Lighting
6. Hellas Sports Lighting / GeoSport
7. IKIO Sports Lighting

C. Substitution Requests: Submit requests for substitutions no later than 10 days prior to bid opening. Include photometric data, warranty details, compliance certifications, and evidence of meeting all performance requirements.

D. Luminaire Characteristics:

1. Comply with UL 1598.
2. Housing: Die cast aluminum with heat sink.
3. Finish: powder coated with high performance polyester paint.
4. Minimum IP65 ingress protection and IK07 impact rating.
5. Mounting Bracket: Adjustable yoke allowing for 360-degree orientation.
6. Exposed Hardware: Stainless-steel latches, fasteners, and hinges.
7. Optics: Arranged to provide various horizontal and vertical beam angles using standard NEMA beam type classifications. Axially symmetrical beam patterns are not acceptable.
8. Correlated Color Temperature (Tk): 5700k.
9. CRI (Ra): ≥ 75.
10. Spill-Light Control Devices: Internal louvers, shields, and external visors furnished by manufacturer and designed for secure attachment to luminaires. Provide as required to comply with obtrusive lighting control requirements.
11. Surge Protection: Provide integral type 2 surge protection devices with peak single impulse surge current rating of 10 kA per phase.

E. LED Driver Characteristics:

1. Comply with UL 8750.
2. Remote mounted in Electrical Component Enclosures.
3. Provide one driver per luminaire.
4. Efficiency: 95 percent minimum.
5. Inrush current: Comply with NEMA 410.
6. Total Harmonic Distortion: maximum 20 percent at full output.
7. Dimming range: 15 to 100 percent. Dim to off is not acceptable.

2.4 POLES AND SUPPORT STRUCTURES

A. Delegated Design:
1. Existing poles and pole foundations shall remain and be reused.
2. Engage a qualified professional engineer, licensed in the jurisdiction where Project is located, to perform a structural analysis of the existing poles and foundations, and to design any necessary modifications, reinforcements, or attachments to support the new luminaires and appurtenances.
3. Structural Characteristics: Comply with the adopted local building code and AASHTO LTS.
4. Dead Load: Weight of new luminaires and their supports, plus existing supporting structure, applied according to AASHTO LTS.
5. Live Load: For service walkways and platforms, apply single load of 500 lbf distributed according to AASHTO LTS.
6. Ice Load: Load of 0 lbf/sq. ft., as applicable for Lewisville, Town of Flower Mound, and the Colony, Texas areas on the Ice Load Map per ASCE 7.
7. Wind Load: Pressure of wind on existing pole, new luminaires, and all other appurtenances, calculated and applied according to AASHTO LTS.
8. Basic wind speed for calculating wind load for poles: Determined from ASCE 7 wind map with a minimum of 150 mph exposure category C.
   a. Minimum Design Life: 50 years.
9. Strength Analysis: For each existing pole, multiply the actual EPA of new luminaires and brackets by a factor of 1.1 to obtain the EPA to be used in pole verification strength analysis. The analysis shall confirm that the existing poles meet or exceed the required strength for the calculated loads.
10. Soil Parameters: Refer to geotechnical report. If no geotechnical report is available, base design on soils that meet or exceed those of a class 5 material as defined in the building code (suitable Lewisville, Town of Flower Mound, Carrolton, and The Colony areas areas).
11. Submittals: Submit a structural analysis report stamped by the professional engineer, including calculations, assumptions, and certification that the existing poles and foundations are structurally sound for reuse with the new lighting system. Identify any required repairs, reinforcements, or modifications.

B. Inspection and Verification:
1. Prior to installation of new luminaires, inspect existing poles for signs of damage, corrosion, cracking, deformation, or other defects that could compromise structural integrity.
2. For steel poles: Perform visual inspection and, if recommended by the engineer, non-destructive testing (e.g., ultrasonic or magnetic particle inspection) to assess wall thickness, weld integrity, and corrosion levels.
3. For prestressed concrete poles: Perform visual inspection and, if recommended by the engineer, non-destructive testing (e.g., impact-echo or ground-penetrating radar) to assess concrete condition, strand integrity, and voids.
4. Verify grounding systems, handholes, and access points are functional and comply with current codes.
5. Repair or reinforce any deficiencies identified in the inspection, subject to approval by the engineer and Owner.

C. Source Limitations: Obtain any new attachments, reinforcements, or appurtenances from a single manufacturer or producer compatible with the existing poles.

D. General Requirements:

1. Shape: Round, tapered (verify existing configuration).
2. Mounting Provisions: Utilize existing direct buried or butt flange for bolted mounting to foundation.
3. Handholes, cable entry holes, and wiring access holes: Minimum clear opening of 3 by 5 inches, with weatherproof cover or adapter plates secured by stainless-steel captive screws. Repair or replace as needed.
4. Cable Support Grip: Wire-mesh type with rotating attachment eye, sized for diameter of cable and rated for a minimum load equal to weight of supported load multiplied by a 5.0 safety factor.

E. Steel Poles (If Applicable to Existing):

1. Comply with ASTM A 572, high strength low alloy steel with a minimum yield of 55,000 psig; verify overlap telescoping joints up to 120 feet in height with access handholes in pole wall.
2. Finish: Verify existing hot-dip galvanize according to ASTM A123; repair any damage to galvanizing.
3. Integral ground lug welded inside the pole opposite the hand hole. Listed for attaching grounding and bonding conductors of type and size indicated. Test and repair as needed.
4. Corrosion Protection: For direct buried steel sections, verify or apply protective epoxy coating to both the pole interior and exterior to 18-inches above ground level.

F. Prestressed Concrete Poles (If Applicable to Existing):

1. Finishing: Verify capped at top and plugged at bottom. Seat each reinforcing strand with epoxy adhesive; inspect for integrity.
2. Grounding: Continuous #6 AWG copper ground wire. Terminate at top of pole and attach to any lightning protection air terminal. Test continuity.
3. Wireway: Smooth, internal, and not less than 3 inches in diameter.
4. Concrete: Minimum 28-day compressive strength of 9500 psi (verify through records or testing if available).
5. Cured with wet steam and aged for a minimum of 15 days prior to installation (historical compliance assumed; verify condition).
6. Internal Reinforcement:
7.
    a. Reinforcing bars complying with ASTM A615.
    b. Cold drawn spiral wire complying with ASTM B240.
    c. Pre-stressing strand, 270 K, 7-wire, complying with ASTM A416.
8. Finish Texture: Standard form.
9. Nameplate: Aluminum cast into pole wall at approximately 5 feet above ground line, listing name of manufacturer, Project identifier, overall height, and approximate weight.

G. Mounting hardware, supports, and fasteners: Size and type as required by luminaire manufacturers' mounting requirements and capable of withstanding 150 mph winds while maintaining luminaire aiming alignment.

1. Mounting Hardware: Stainless steel, 18-8 grade or better, passivated and coated with aluminum based thermosetting epoxy resin for protection against corrosion and stress corrosion cracking.
2. Mounting Supports: Galvanized steel meeting ASTM A153 for hot dipped galvanizing, ASTM B695 for mechanical galvanizing, or other high-strength corrosion-resistant material.
3. Fasteners: Compatible with poles as well as the substrates to which poles and standards are fastened and shall not cause galvanic action at contact points.

H. Lightning Protection: Verify existing, otherwise, provide lightning protection for the existing poles and pole foundations in accordance with NFPA 780. Provide components listed per UL 96 and UL 96A.

1. Size down conductors and grounding electrode conductors in accordance with NFPA 780.
2. Grounding Electrodes: Provide ground rods where integrated concrete encased grounding electrodes are not provided with pole base.
   a. Ground Rod: 10 mil pure electrolytic copper coating with molecular bond to high strength steel core; 3/4 inch by 10 feet with chamfered end.

2.5 POWER DISTRIBUTION AND CONTROL

A. Electric Power: 480 V, three phase unless indicated otherwise on drawings.

1. Existing electrical circuits from the source breaker to the pole shall remain and be reused.
2. The source existing 3-pole 480V breaker varying in size from 100A to 225A may be replaced if necessary based on the specific manufacturer's requirements for the new lighting system. Bidder shall account for all such changes, including any associated wiring modifications, overcurrent protection adjustments, and coordination with existing systems in the bid.
3. Balance load between phases. Install wiring to balance three phases at each support structure.
4. Include code required overcurrent protective devices, verifying compatibility with existing circuits.
5. Maximum Total Voltage Drop from Source to Load: 3 percent, including voltage drops in branch circuit, subfeeder, and feeder. Verify existing circuits meet this requirement or modify as needed.

B. Inspection and Verification of Existing Electrical Systems:

1. Prior to installation of new components, inspect existing electrical circuits, wiring, breakers, and enclosures for signs of damage, corrosion, insulation degradation, loose connections, or other defects that could compromise safety or performance.
2. Perform electrical testing, including continuity, insulation resistance, and ground fault checks, as recommended by a qualified electrician or engineer.
3. Verify compliance with current National Electrical Code (NEC) and local amendments in Lewisville, Town of Flower Mound, and The Colony, Texas.

4. Repair or replace any deficiencies identified in the inspection, subject to approval by the engineer and Owner. Bidder shall include costs for anticipated repairs in the bid.

C. Pole Harness: Provide hard wired, unspliced conductors terminated with quick connect plugs, abrasion protection sleeve, and strain relief for full pole height. Intermediate splices or connectors are not acceptable.

D. Electrical Cabinets and Enclosures Exposed to Weather: NEMA 250, Type 4 enclosure constructed from corrosion-resistant material, with stainless steel hinges and lockable latches. Inspect and repair or replace existing enclosures as necessary.

E. Surge Protection: Provide type 2 surge protection at each pole enclosure and at each control cabinet.
   1. Comply with UL 1449.
   2. Digital indicator lights.
   3. Peak Surge Current Rating per Phase: equal to or greater than 40 kA for each line to ground as recommended by IEEE C62.41.2.
   4. Replace existing surge protection devices to meet this requirement

F. Pole Electrical Component Enclosures: contains equipment required to operate luminaires. Mount to pole at 10 feet above grade for easy servicing. Existing enclosures in general shall be removed and provide new as needed.
   1. Disconnect: one per circuit
   2. Terminal Blocks
   3. Ground bar
   4. Surge Protection
   5. Fuses: Individual fuses for each driver. Provide spare fuses.
      a. Circuit breakers in lieu of fuses are acceptable.
   6. Remote Drivers: up to four per enclosure.
   7. Controller

G. Lighting Control: digital, programmable; providing the following functions:
   1. System controls shall be mobile app capable, with a dedicated mobile application required for full system operation, compatible with iOS (e.g., iPhone) and Android devices. This is a mandatory requirement and not optional.
   2. Provide for instant on/off and full dimming with addressable control of individual luminaires.
   3. Contactor Cabinet: Enclosure with electrically operated, dual acting single solenoid mechanism, mechanically held in both open and closed positions, combination-type remote control lighting contactors, complying with NEMA ICS 2 and UL 508.
      a. Current Rating for Switching: 30A up to 600VAC.
      b. Fault Current Withstand Rating: Equal to or exceeding the available fault current at the point of installation.
      c. Visual position indicator.
      d. Control Coil Voltage: Match control power source.
      e. Provide contactor capacity and configuration as required to accommodate switched zones shown on drawings plus 4 spares.
      f. Provide Hand-off-auto (HOA) selector switch, of the heavy-duty “oil-tight”, maintained-contact type, mounted on the front cover with legend plate for each switched zone.

4. Control and Monitoring:
    a. Scheduling: Accept and store 7-day schedules with curfew limits.
    b. Incorporate non-volatile memory. Should power be interrupted and subsequently restored, control settings and parameters saved in protected memory shall not be lost and should restore system to its last operating state without requiring user input.
    c. Security: Multi-level password security designed to restrict user access. Provide the following minimum access levels:
        1) User Level: Modifiable password that only allows early off capabilities.
        2) Advanced: allows full scheduling capabilities.
        3) Service: allows users (authorized services technicians only) access to advanced and service tools.
    d. Remote Control: Allow users with a security code to remotely schedule and control on/off system operation via a web site, phone, email, and required mobile app up to ten years in advance.
        1) Provide Digital Cellular Antenna with two-way TCP/IP communication link.
        2) Mobile app shall be compatible with iOS devices (e.g., iPhone) and Android, offering intuitive interface for control and monitoring.
    e. Press Box Integration: Provide capability for remote scheduling and control via mobile app or tablet directly from the press box, ensuring intuitive operation of RGBW presets during game events
    f. Remote Monitoring and System Management Tools: Monitor lighting performance, hours of usage, system temperature, switch position, and contactor status through a web-based dashboard interface. Capable of notifying owner and manufacturer of an individual luminaire outage or predictive maintenance threshold is exceeded so appropriate maintenance can be scheduled.
        1) The dashboard application shall be accessible through mobile applications suitable for iOS and Android mobile devices.
        2) Include all infrastructure and annual costs for operating the remote monitoring and management tools for the duration of the system warranty and maintenance agreement.
    g. Master control station: Key-operated, manual push-button controls, and system status indicator lights. Provide adjacent to the control cabinet to allow mode selection.
    h. Provide preprogrammed illuminance modes with the following as a minimum:
        1) Full Match Mode (High): operating in a manner that satisfies competition performance requirements
        2) Training Mode (Medium): Operate with an average horizontal illuminance reduced to 75 percent of competition performance requirements.
        3) Maintenance Mode (Low): Operate with an average horizontal illuminance reduced to 50 percent of competition performance requirements.
        4) Emergency Egress Mode: Activated when primary power source fails. Provide illumination required for player and spectator egress.
        5) Custom Mode: Additional mode for Owner-defined settings.

i. Entertainment Controls: Provide supplemental show control components that allow users to select predesigned special effects through a touchscreen controller.
   1) Include theatrical RGB controls for show lights, with at least 5 presets (to be determined by Owner) for color-changing effects, sequencing, and dynamic lighting scenes.
   2) Include the initial scene setup and an addition 20 hours of special effects programming in the base bid.

2.6 EMERGENCY EGRESS LIGHTING

A. General: Provide emergency egress lighting for the bleachers (home and visitor) complying with the International Building Code (IBC) photometric criteria. Emergency lighting shall activate automatically upon failure of the normal power supply, providing illumination within 10 seconds.

1. Illuminance Criteria: Minimum average of 1.0 fc, minimum of 0.1 fc, with a maximum-to-minimum ratio greater than 40:1, measured at the walking surface along designated spectator egress pathways on the bleachers.
2. Dedicated Fixtures: Provide a minimum of one luminaire per sports lighting pole dedicated to emergency egress illumination. Luminaires shall be integrated with the athletic field lighting system and aimed to cover egress pathways.

B. Emergency Power Source: Provide outdoor rated emergency inverter similar to **IOTA IIS EXT 1750 Emergency Inverter by Acuity Brand**, configured with voltages compatible with the 480V three-phase lighting system (or 277V as compatible with the selected field lighting manufacturer's system).

1. Capacity: Sufficient to power dedicated emergency luminaires for a minimum runtime of 90 minutes.
   a. Submit calculations for review by engineer.
2. Battery: Lead-acid type, sealed and maintenance-free. Other battery types (such as lithium) will be considered provided they meet the environmental rating requirements for the specific site location.
3. Mounting: Pole-mounted near the base of the pole in a weatherproof NEMA 3R enclosure.
4. Dimensions: Physical size of inverter shall be appropriate for location at base. It shall not extrude such that it impedes the flow of traffic. Dimensions are subject to approval by engineer.
5. Environmental Ratings: Rated for outdoor use with operating temperature range of **20°F to 110°F**; include internal heater and temperature-controlled forced air cooling as required to achieve this range for site conditions in Lewisville ISD locations.
6. Integration: Battery backup system shall be fully integrated with the lighting control system, including automatic transfer upon power failure and return to normal operation upon restoration. Include self-diagnostic features, test switches, and status indicators.
7. Compliance: UL 924 listed for emergency lighting equipment. Provide photometric calculations demonstrating compliance with IBC egress criteria as part of delegated design submittals.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine roughing-in for luminaire electrical and communications conduit to verify actual locations of connections luminaire installation. Verify compatibility of existing conduits and wiring with new luminaires

C. Examine existing soils and foundations for suitable conditions where luminaires will be installed. Notify Architect and Engineer if soil conditions differ from basis of existing foundation design or if deficiencies are observed.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 FOUNDATIONS

1. Existing foundations shall remain and be reused. Verify integrity as part of Delegated Design requirements in Section 1.1 Poles and Support.

3.3 INSTALLATION

A. Wiring Method: Install cables and conductors in raceways, except when cables are enclosed within boxes, poles, and support structures. Conceal raceways and cables. Exposed conductors or cabling are not permitted. Comply with requirements in Section 260519, Low-Voltage Electrical Power Conductors and Cables and Section 260533, Raceways and Boxes for Electrical Systems. Reuse existing wiring where possible, verifying condition and capacity.

1. Minimum lighting branch circuit conductor size: 8 AWG copper. Verify or replace existing conductors as needed to comply.
2. Ensure cables from luminaires at top of pole to control box are well supported, without strain on connections.

B. Utilize silicone filled water resistant connectors for all outdoor connections

C. Pole Installation

1. Use web fabric slings (not chain or cable) to raise and set structural support members. Protect equipment during installation to prevent corrosion.
2. Verify existing poles and other structural units level, plumb, and square.
3. Alignment: Align existing poles and pole foundations for optimum directional alignment of luminaires and their mounting provisions on pole.
4. Except for embedded structural members, grout void between pole base and foundation if gaps exist. Use nonshrinking or expanding concrete grout firmly packed in entire void space. Use a short piece of 1/2-inch- diameter pipe to make a drain hole through grout. Arrange to drain condensation from interior of pole.

D. Install luminaires at height and aiming angle as necessary based on Delegated Design Requirements.

1. Factory pre-aimed systems are acceptable for initial installation, but final fixture aiming shall be conducted using a laser mounted to the luminaire to point the fixture at precise computer generated aiming points on the playing surface using (X,Y) coordinates.

E. Install control devices and remote drivers in cabinets mounted on support structure at least 10 feet above finished grade. If field conditions require deviation from this, they shall be taken into consideration.

F. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.

G. Where a tightening torque is indicated as a numeric value on equipment or in installation instructions provided by the manufacturer, use a calibrated torque tool to achieve that indicated torque value, unless the equipment manufacturer has provided installation instructions for an alternative method of achieving the required torque.

3.4 GROUNDING

A. Ground Metal Poles and Support Structures: Comply with requirements in Section 260526 "Grounding and Bonding for Electrical Systems." Verify existing grounding systems and repair or enhance as needed

1. Install grounding electrode for each pole unless otherwise indicated, or verify existing.
2. Install grounding conductor pigtail in the base for connecting luminaire to grounding system, or verify existing.

3.5 IDENTIFICATION

A. Identify system components, wiring, cabling, and terminals. Comply with requirements for identification specified in Section 260553 "Identification for Electrical Systems."

1. Label each luminaire with column/pole location and a unique Luminaire number matching the identifier in the control software.
2. Label all control components and switches to match field diagrams.
3. Provide directories inside contactor cabinets that identify each relay and the associated control zone.
4. Provide updated directories inside the existing switchboards/ distribution panels after all modifications.

3.6 FIELD QUALITY CONTROL

A. Perform the following tests and inspections with the assistance of a factory-authorized service representative. **All field quality control testing and manufacturer verification**

shall be completed and any deficiencies corrected prior to issuance of the final punch list for each school.

B. Provide written 7-day advance notice of any testing to allow The Engineer and Owner's Representative the option to witness the tests

C. Acceptance Tests:

1. After installing sports lighting system and after electrical circuits have been energized, perform proof-of-performance field measurements and analysis for compliance with requirements.
2. Conduct illumination measurements in accordance with IESNA LM-5-04.
3. Playing and Other Designated Areas: Make field measurements at intersections of grids, dimensioned, and located as specified in "Performance Requirements" Article
4. Make field measurements at established test points in areas of concern for spill light and glare.
5. Perform analysis to demonstrate correlation of field measurements with specified illumination quality and quantity values and corresponding computer-generated values that were submitted with engineered design documents. Submit a report of the analysis. For computer-generated values, use manufacturer's lamp lumens that are adjusted to lamp age at time of field testing.
6. **The lighting system manufacturer's authorized technical representative shall make a physical site visit to each school after all installation and energization is complete but prior to final punch list or Substantial Completion. The representative shall personally witness, verify, and confirm in writing the authenticity and accuracy of the as-installed photometric performance measurements against the approved design submittals and computer calculations. The representative shall certify compliance in a signed letter submitted to the Owner and Engineer before final punch. Any discrepancies shall be corrected prior to final acceptance.**

D. Correction of Illumination Deficiencies for Playing Areas: Make corrections to illumination quality or quantity, measured in field quality-control tests, that varies from specified illumination criteria.

1. Add or replace luminaires; change mounting height and aiming; or install louvers, shields, or baffles.
2. If luminaires are added or mounting height is changed, revise aiming and recalculate and modify or replace support structures if indicated.
3. Do not replace luminaires with units of higher wattage without Engineer's approval.
4. Retest as specified above after repairs, adjustments, or replacements are made.
5. Report results in writing.

E. Correction of Excessive Illumination in Spill-Light-Critical Areas: If measurements indicate that specified limits for spill light are exceeded, make corrections to illumination quantity, measured in field quality-control tests, that reduce levels to within specified maximum values.

1. Replace luminaires; change mounting heights and revise aiming; or install louvers, shields, or baffles.
2. Obtain Engineer's approval to replace luminaires with units of higher wattage.

3. If mounting height is changed, revise aiming and recalculate and modify or replace support structures if indicated.
4. Retest as specified above after repairs, adjustments, or replacements are made.
5. Report results in writing.

F. Test and Inspection Reports: Prepare a written report to certify compliance with test parameters and record the following:

1. Test procedures used.
2. Results that comply with requirements.
3. Results that do not comply with requirements and corrective action taken to achieve compliance with requirements.

G. System will be considered defective if it does not pass tests and inspections. Remove and replace malfunctioning units and retest as specified above.

H. Drone Documentation: Contractor shall provide before and after drone footage of **each school,** the existing and new lighting systems, included in the base bid price.

1. Before Installation: Drone footage of existing lighting during the day and after dark, capturing overall field illumination, uniformity, and any visible issues.
2. After Installation: Drone footage of new lighting during the day and after dark, demonstrating improved illumination, uniformity, and compliance with performance requirements.
3. Footage Requirements: High-resolution video (minimum 4K), aerial views from multiple angles and altitudes, narrated or annotated to highlight key differences. Submit digital files to Owner and Engineer within 30 days of substantial completion.

3.7 ADJUSTING

A. Post-Occupancy Adjustments: When requested within 12 months of date of Completion, provide on-site assistance in adjusting aimable luminaires to suit actual occupied conditions. Provide up to two visits to Project during other-than-normal occupancy hours for this purpose. Some of this work may be required after dark.

3.8 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's personnel to adjust, operate, and maintain systems.

B. Training shall consist of not less than 4 hours **per school** of onsite hands-on instruction for a minimum of four attendees **per school.** Training shall cover the following topics at a minimum:

1. Theory of operation.
2. Operation of system and components, including app-based controls via mobile devices (e.g., iPhone).
3. Maintenance and configuration, including RGB theatrical controls and preset management.
4. Troubleshooting and Repair.
5. Replacement of components.

C. Lighting Controls Manufacturer/Vendor Onsite Meeting: In addition to training, the lighting controls manufacturer or vendor shall provide one (1) dedicated onsite meeting per school with Owner's personnel. The meeting shall cover full system operation, mobile app functionality (iOS and Android), RGB theatrical show controls and presets, scheduling, monitoring, troubleshooting, and allow hands-on questions. The meeting shall be scheduled at a mutually agreeable time prior to Final Completion for each school.

END OF SECTION